Maximality of the Došen–Petrić Boolean Category

Florian Lengyel*

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Abstract

We prove that the zero-identity Boolean category of Došen and Petrić, freely generated by a nonempty set of propositional variables, is maximal. Adjoining any equation between parallel arrows that does not follow from the defining equations, and closing under the arrow operations and uniform formula substitution, collapses this category to a preorder. This disproves their conjecture that the category is not maximal. The proof uses Boolean Coherence to select a pair of variable occurrences on which the two arrows differ. Composition with arrows selecting these occurrences, uniform substitution and two isomorphisms yield equality of the product projections. We prove that any zero-identity Boolean category satisfying such an additional equation under every interpretation of the variables is a preorder. We also give a Lean formalization of these proofs and their dependencies. It includes Boolean Coherence and the correspondence between substitution on tagged occurrence sets and substitution on finite ordinals.

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1 Introduction

We prove that the freely generated zero-identity Boolean category of Došen and Petrić over a nonempty set of propositional variables is maximal. Adding any equation between parallel arrows that does not follow from its defining equations, and closing under the arrow operations and uniform formula substitution, collapses the category to a preorder. This disproves Došen and Petrić's conjecture that the category is not maximal [1, §14.2, p. 322]. Before adjoining the equation, the two projections $p \wedge p \to p$ are distinct for every propositional variable p.

The proof reduces an added equation to equality of the product projections, as in the maximality theorem for cartesian categories [2, §5]. Boolean Coherence supplies a pair of variable occurrences belonging to one arrow's relation and not the other. Precomposition and postcomposition isolate this pair and give the equation of [1, §9.7, pp. 215–216]. Uniform substitution and the double-negation isomorphism then give this equation for every formula. The two isomorphisms listed in [1, §12.5, p. 294] give equality of the product projections. Sections 4 and 5 present this proof; Section 6 gives a second proof using mix. The proof of [3, Lemma 7] also supplies the two isomorphisms.

Other notions of Boolean category need not make conjunction and disjunction categorical products and coproducts. Free categories constructed from classical proof nets are studied in [4, Theorem 3.5.4], equations on monoid and comonoid structures in $*$-autonomous categories in [5],

*Email: flengyel@gradcenter.cuny.edu

and inequalities interpreting cut reduction in [6]. Section 2 compares the negation equation of [6] with the zero-identity equation used here.

We use the occurrence coordinates of [7, §9]: occurrences form tagged disjoint unions, and substitution replaces each occurrence by a block. Section 3 defines these coordinates; Appendix A proves their correspondence with the finite-ordinal substitution formula of Došen and Petrić. Section 6 and Appendix D describe the Lean formalization, including a guide to its declarations and proof steps. Appendix E constructs the two inverses used in the first maximality proof, and Appendix C corrects two typographical errors in the cut-elimination argument used in Došen and Petrić's proof of Boolean Coherence [1]. References to [1] use the printed pagination of its September 2007 revised version. Lean declaration names in footnotes omit the namespace prefix `DosenPetric.`

2 The Boolean proof category

Fix a nonempty set P of propositional variables. Write $\mathrm{Form}(P)$ for the formulas built from P, the constants $\top, \bot$, the binary connectives $\wedge, \vee$, and negation $\neg$. Let $\mathsf{B}(P)$ be the freely generated zero-identity Boolean category defined in [1, §§14.2–14.3]. Its objects are these formulas, and its arrows are the arrow terms of that presentation modulo its equations. We write B when P is fixed. We recall the structure and equations used in the proof.

The connectives $\wedge$ and $\vee$ give binary products and coproducts; $\top$ is terminal and $\bot$ is initial. For $A_1 = A$ and $A_2 = B$, denote the projections and injections by

$$\pi_i^{A,B} : A \wedge B \longrightarrow A_i, \qquad \iota_i^{A,B} : A_i \longrightarrow A \vee B \qquad (i = 1, 2).$$

For $f : X \to A$ and $g : X \to B$, write $\langle f, g\rangle : X \to A \wedge B$ for their pairing. For $f : A \to X$ and $g : B \to X$, write $[f, g] : A \vee B \to X$ for their copairing. Write $!_A : A \to \top$ and $\kappa_A : \bot \to A$ for the terminal and initial arrows, and $1_A : A \to A$ for the identity. Composition $g \circ f$ applies f first. For $f : A \to A'$ and $g : C \to C'$, the product and coproduct bifunctors give arrows

$$f \wedge g : A \wedge C \longrightarrow A' \wedge C', \qquad f \vee g : A \vee C \longrightarrow A' \vee C'.$$

The dissociativity arrow is

$$d_{A,B,C} : A \wedge (B \vee C) \longrightarrow (A \wedge B) \vee C.$$

Its first-injection equation is

$$d_{A,B,C} \circ (1_A \wedge \iota_1^{B,C}) = \iota_1^{A \wedge B, C}. \tag{1}$$

This is the coproduct-injection equation of [1, §11.1, p. 234].

Negation is represented by a functor $\neg : \mathsf{B}^{\mathrm{op}} \to \mathsf{B}$, sending $f : A \to C$ to $\neg f : \neg C \to \neg A$. Its double-negation and De Morgan isomorphisms are natural. In particular, write $n_A : \neg\neg A \to A$ for the double-negation isomorphism. The presentation also has arrows $\eta_A : \top \to \neg A \vee A$ and $\epsilon_A : A \wedge \neg A \to \bot$. The *zero identity* is the endomorphism $0_A : A \to A$ defined by

$$0_A = [\kappa_A, 1_A] \circ (\epsilon_A \vee 1_A) \circ d_{A,\neg A,A} \circ (1_A \wedge \eta_A) \circ \langle 1_A, !_A\rangle. \tag{2}$$

The defining zero-identity equation asserts that

$$f \circ 0_A = 0_C \circ g \qquad (f, g : A \to C). \tag{3}$$

Equation (2) is $(0\eta\epsilon)$ of [1, §14.2, p. 317]. Its product and coproduct unit isomorphisms are expressed here in pairing and copairing notation. Equation (3) is the zero-identity law $(0I)$; see [1, §14.3, p. 322].

Appendix B displays equations (1)–(3) as commutative diagrams.

Whenever $\mathsf{B}(A, C)$ is nonempty, define $0_{A,C} = f \circ 0_A$ for any $f : A \to C$. Equation (3) makes this independent of f: both $f \circ 0_A$ and $g \circ 0_A$ equal $0_C \circ g$. Taking $f = g$ gives $f \circ 0_A = 0_C \circ f$. Consequently zero arrows absorb composition whenever the composites are defined: $h \circ 0_{A,C} = 0_{A,D}$ for $h : C \to D$, and $0_{A,C} \circ k = 0_{X,C}$ for $k : X \to A$.

Führmann and Pym interpret cut reduction by inequalities between arrows. An inequality belongs to a net theory if and only if it holds in every interpretation of that theory in their classical categories [6, Theorems 6.4–6.5]. Their negation triangle makes the composite in (2), with the corresponding monoidal unit isomorphisms, equal to 1_A [6, (11)]. Here that composite is 0_A.

Two arrows are *parallel* when they have the same source and target. A *proof congruence* on B is an equivalence relation on each hom-set preserved by composition and every arrow operation of the presentation, including the derived operations of pairing, copairing, and negation. A *substitution congruence* is a proof congruence also preserved by every uniform substitution $\sigma : P \to \mathrm{Form}(P)$. Here substitution acts on the entire arrow term, including its source and target; this is rule (su) of [1, §2.7, p. 51].

The *total congruence* identifies every parallel pair. A category is *thin*, or a *preorder*, when each hom-set has at most one element. For a proof congruence θ, the quotient B/θ has the same objects as B and arrows given by θ-equivalence classes. It is a preorder exactly when θ is total.

Theorem 2.1. *Let P be a nonempty set of propositional variables, and let $\mathsf{B}(P)$ be the freely generated zero-identity Boolean category of Došen and Petrić. For every substitution congruence θ on $\mathsf{B}(P)$ that identifies two distinct parallel arrows, the quotient $\mathsf{B}(P)/\theta$ is a preorder.*

Maximality means that this collapse occurs for every added equation between parallel arrow terms that does not follow from the defining equations, with closure under the arrow operations and uniform substitution. Equivalently, the only substitution congruences are equality and the total congruence. Sections 4 and 5 give the proof.

3 Occurrences and substitution

For a formula A, write $\mathrm{Occ}(A)$ for the finite set of positions occupied by propositional variables in A. Repeated occurrences of the same variable occupy distinct positions. We use disjoint unions to distinguish these positions.

For finite sets X, Y, let $X \sqcup Y = (\{1\} \times X) \cup (\{2\} \times Y)$ be their tagged disjoint union. The occurrence sets are defined recursively by

$$\begin{aligned} \mathrm{Occ}(p) &= \{*\}, & \mathrm{Occ}(\top) = \mathrm{Occ}(\bot) &= \varnothing, \\ \mathrm{Occ}(\neg A) &= \mathrm{Occ}(A), & \mathrm{Occ}(A \star C) &= \mathrm{Occ}(A) \sqcup \mathrm{Occ}(C) \quad (\star \in \{\wedge, \vee\}). \end{aligned}$$

The label map $\lambda_A : \mathrm{Occ}(A) \to P$ assigns to each occurrence its underlying variable. Distinct positions may have the same label. Numbering the occurrences from left to right gives a bijection $\mathrm{Occ}(A) \to \{1, \ldots, |\,\mathrm{Occ}(A)|\}$; the latter set is empty when $|\,\mathrm{Occ}(A)| = 0$. Appendix A.2 gives the substitution formula in these finite-ordinal coordinates.

Let $\mathbf{Rel}_{\mathrm{fin}}$ be the category of finite sets and binary relations. A relation $R : X \to Y$ is a subset of $X \times Y$; identities are diagonal relations, and composition is relational composition. Transporting the occurrence interpretation of [1] to the tagged coordinates gives a functor $G : \mathsf{B} \to \mathbf{Rel}_{\mathrm{fin}}$ with $GA = \mathrm{Occ}(A)$. For $f : A \to C$, its image is the relation $Gf \subseteq \mathrm{Occ}(A) \times \mathrm{Occ}(C)$. For example,

$$G\pi_1^{A,C} = \{((1, x), x) : x \in \mathrm{Occ}(A)\}, \qquad G\iota_1^{A,C} = \{(x, (1, x)) : x \in \mathrm{Occ}(A)\}.$$

The second projection and injection use tag 2. Pairing takes the union of the two relations after tagging their targets; copairing takes the union after tagging their sources. The relation $Gd_{A,B,C}$ is the graph of the rebracketing bijection

$$\mathrm{Occ}(A) \sqcup (\mathrm{Occ}(B) \sqcup \mathrm{Occ}(C)) \longrightarrow (\mathrm{Occ}(A) \sqcup \mathrm{Occ}(B)) \sqcup \mathrm{Occ}(C),$$

which preserves each occurrence of A, B, C [1, §2.9, p. 60]. Zero arrows have empty relations.

For an occurrence $x \in \mathrm{Occ}(A)$, define $\mathrm{sgn}_A(x) = (-1)^k \in \{+1, -1\}$, where k is the number of negations above x in A. Every arrow $f : A \to C$ preserves both the variable and its sign:

$$(x, y) \in Gf \quad \Longrightarrow \quad \lambda_A(x) = \lambda_C(y), \qquad \mathrm{sgn}_A(x) = \mathrm{sgn}_C(y).$$

Induction on arrow terms proves this assertion. Identities, associativity and symmetry maps, projections and injections, diagonals and codiagonals, mix, dissociativity and unit maps preserve the variable and its sign. The double-negation and De Morgan maps preserve the signs, and the zero, η and ϵ maps have empty relations. Composition and the two binary operations preserve the assertion.[1]

Theorem 3.1 (Boolean Coherence, Došen–Petrić)**.** *The functor $G : \mathsf{B} \to \mathbf{Rel}_{\mathrm{fin}}$ is faithful: parallel arrows f, g of B are equal if and only if $Gf = Gg$.*

This is [1, §14.2, p. 321]. In particular, for $p \in P$, the two projections $p \wedge p \to p$ have different occurrence relations, so B is nonthin.

Joyal's collapse result says that a cartesian closed category with an initial object $\bot$ is a preorder if $A \cong (A \Rightarrow \bot) \Rightarrow \bot$ for every object A [1, §14.3, p. 324]. The category B is not cartesian closed. Faithfulness gives at most one arrow $\top \to E$ for every E, since $\mathrm{Occ}(\top)$ is empty, whereas $1_p \neq 0_p$ for $p \in P$. An exponential $p \Rightarrow p$ would give a bijection $\mathsf{B}(p, p) \cong \mathsf{B}(\top, p \Rightarrow p)$, a contradiction.

The maximality theorem for cartesian closed categories [8, §11] uses a typed Böhm theorem. The absence of $p \Rightarrow p$ prevents its application here.

A *literal* is p or $\neg p$ for $p \in P$. A formula is in *negation normal form* when negation occurs only immediately before a propositional variable. The double-negation and De Morgan isomorphisms, including the isomorphisms for the truth constants, give an isomorphism $a_A : A \to A^\circ$ to a formula in negation normal form. The relation Ga_A is the graph of a bijection $\mathrm{Occ}(A) \to \mathrm{Occ}(A^\circ)$ [1, §14.1, pp. 313–315].

For arrows between formulas in negation normal form, every pair in the occurrence relation consists of occurrences of the same literal. In the presentation on negation normal forms in [1, §14.2, pp. 318–321], objects are formulas built from literals, conjunction, disjunction and the truth constants. Arrow terms are generated by identities, associativity and symmetry maps, diagonals and codiagonals, projections and injections, mix, dissociativity, unit maps, zero endomorphisms, and η_p, ϵ_p. The zero endomorphisms and η_p, ϵ_p have empty relations; each remaining generator joins equal literals. Composition, $\wedge$ and $\vee$ preserve this property. The equivalence with B preserves the occurrence interpretation, so the assertion follows.

Negation takes the converse relation:

$$(y, x) \in G(\neg f) \quad \Longleftrightarrow \quad (x, y) \in Gf \qquad (f : A \to C).$$

On the normal-form presentation, negation interchanges the two binary operations and reverses each generator relation and the order of composition. Induction on terms therefore gives the converse

[1] Lean: `Presentation.occurrence_signedLabel`.

relation. The occurrence bijections of normalization transport this equation to the original positions, using $\mathrm{Occ}(\neg A) = \mathrm{Occ}(A)$ and $\mathrm{Occ}(\neg C) = \mathrm{Occ}(C)$.[2]

Substitution replaces each occurrence of a variable by a separate block of occurrences of its substituted formula. Appendix A.1 describes these blocks and their label maps by commutative diagrams, following [7, §9]. Let Q be a set of propositional variables and $\sigma : P \to \mathrm{Form}(Q)$ a substitution. If $A[\sigma]$ denotes the result of uniform substitution, there is a bijection

$$\mathrm{Occ}(A[\sigma]) \cong D_{A,\sigma}, \qquad D_{A,\sigma} = \coprod_{x \in \mathrm{Occ}(A)} \mathrm{Occ}(\sigma(\lambda_A(x))). \tag{4}$$

The set on the right consists of pairs (x, t), where x is an occurrence of A and t is an occurrence within the formula inserted there. Induction on A proves the bijection: the variable case has one block, the constant cases have none, binary connectives form tagged unions, and negation preserves the occurrence set. In particular, two occurrences of the same variable give two separately tagged copies of its substituted formula.

Put $E_\sigma = \coprod_{p \in P} \mathrm{Occ}(\sigma(p))$. Define $\mathrm{pr}_D : D_{A,\sigma} \to \mathrm{Occ}(A)$ and $\mathrm{pr}_E : E_\sigma \to P$ by $\mathrm{pr}_D(x, t) = x$ and $\mathrm{pr}_E(p, t) = p$. The function $\widehat{\lambda}_{A,\sigma} : D_{A,\sigma} \to E_\sigma$ sends (x, t) to $(\lambda_A(x), t)$. Thus

$$\begin{array}{ccc} D_{A,\sigma} & \xrightarrow{\widehat{\lambda}_{A,\sigma}} & E_\sigma \\ {\scriptstyle \mathrm{pr}_D}\downarrow & & \downarrow{\scriptstyle \mathrm{pr}_E} \\ \mathrm{Occ}(A) & \xrightarrow[\lambda_A]{} & P. \end{array} \tag{5}$$

Both composites send (x, t) to $\lambda_A(x)$. For $(x, t), (x', t) \in D_{A,\sigma}$, their images under $\widehat{\lambda}_{A,\sigma}$ agree exactly when $\lambda_A(x) = \lambda_A(x')$, even if $x \neq x'$.

The same presentation defines $\mathsf{B}(Q)$ when Q is empty. Substitution on arrow terms induces a functor

$$F_\sigma : \mathsf{B}(P) \longrightarrow \mathsf{B}(Q), \qquad A \longmapsto A[\sigma], \quad f \longmapsto f[\sigma].$$

Here $f : A \to C$ is sent to $f[\sigma] : A[\sigma] \to C[\sigma]$. Rule (su) makes this functor well defined on arrow terms modulo the defining equations. It preserves identities, composition, and the specified arrow operations. For each arrow, the blocks in (4) describe the occurrence sets of its substituted source and target.

Write $\chi_{A,\sigma} : \mathrm{Occ}(A[\sigma]) \to D_{A,\sigma}$ for the bijection in (4). For $f : A \to C$, define the relation $R_{f,\sigma} : D_{A,\sigma} \to D_{C,\sigma}$ by

$$\begin{aligned} ((x, t), (y, u)) \in R_{f,\sigma} \quad \Longleftrightarrow \quad & (x, y) \in Gf \ \text{ and} \\ & (\lambda_A(x), t) = (\lambda_C(y), u) \text{ in } E_\sigma. \end{aligned}$$

Thus an edge of Gf is replaced by the identity relation within the inserted block. The substituted relation satisfies

$$(x', y') \in G(f[\sigma]) \quad \Longleftrightarrow \quad (\chi_{A,\sigma}(x'), \chi_{C,\sigma}(y')) \in R_{f,\sigma}.$$

Regard the two bijections as their graphs. The following square then commutes in $\mathbf{Rel}_{\mathrm{fin}}$:

$$\begin{array}{ccc} \mathrm{Occ}(A[\sigma]) & \xrightarrow{G(f[\sigma])} & \mathrm{Occ}(C[\sigma]) \\ {\scriptstyle \chi_{A,\sigma}}\downarrow & & \downarrow{\scriptstyle \chi_{C,\sigma}} \\ D_{A,\sigma} & \xrightarrow[R_{f,\sigma}]{} & D_{C,\sigma}. \end{array}$$

[2] Lean: `Reduced.Term.dual_relation` and `Presentation.occurrence_negate`.

Induction on the arrow term proves the relation formula. Identities, associativity and symmetry maps, projections and injections, diagonals and codiagonals, mix, dissociativity and unit maps act on the inserted blocks by the same tagged relations. The double-negation and De Morgan maps give their occurrence bijections on these blocks; zero, η and ϵ give empty relations. The two binary operations retain the tags. For composition, an intermediate old occurrence supplies an intermediate block, and preservation of variable labels identifies its inner occurrence on both sides. The formula passes to quotient arrows because both interpretations respect the defining equations.[3] Repeated old occurrences retain their different first coordinates; an empty inserted block supplies no positions.

Let R be a set of variables and let $\tau : Q \to \mathrm{Form}(R)$ be another substitution. Define $\tau \star \sigma : P \to \mathrm{Form}(R)$ by $(\tau \star \sigma)(p) = \sigma(p)[\tau]$. For formulas A, C over P and $f : A \to C$, substitution satisfies

$$A[p \mapsto p] = A, \quad (A[\sigma])[\tau] = A[\tau \star \sigma],$$
$$f[p \mapsto p] = f, \quad (f[\sigma])[\tau] = f[\tau \star \sigma].$$

The arrow equalities use the displayed formula equalities to identify their sources and targets. Pointwise equal substitutions give equal substituted formulas and arrows with these same identifications. Formula induction proves the object equations; term induction and passage to the quotient prove the arrow equations.[4]

4 Isolating an occurrence

Put $j_A = \iota_1^{A,\top} : A \to A \vee \top$. We shall derive the equation

$$j_A \circ \pi_1^{A,\bot} = 0_{A\wedge\bot, A\vee\top}. \tag{K_A}$$

The right-hand side is also $\iota_2^{A,\top} \circ !_\bot \circ \pi_2^{A,\bot}$: its occurrence relation is empty, so Boolean Coherence identifies it with the zero arrow. Thus (K_A) is the equation used in [1, §9.7, pp. 215–216].

For negation-free formulas A, B such that $A \to B$ is a tautology, [3, Lemma 6] constructs an arrow whose occurrence relation consists of the edge between any prescribed occurrences of the same variable in A and B. The following lemma isolates an edge of either of two given arrows; normalization allows the formulas to contain negation.

Selecting a pair on which relations differ also occurs in the maximality proofs for presentations of split preorders, split equivalence relations and binary relations [9, §16].

Lemma 4.1 (Isolation of an occurrence)**.** *For distinct parallel arrows $f, g : A \to C$ in B, there are a literal ℓ and arrows*

$$i : \ell \wedge \bot \longrightarrow A, \qquad r : C \longrightarrow \ell \vee \top$$

such that, after possibly interchanging f and g,

$$r \circ f \circ i = j_\ell \circ \pi_1^{\ell,\bot}, \qquad r \circ g \circ i = 0_{\ell\wedge\bot, \ell\vee\top}.$$

Proof. First replace f, g by $a_C \circ f \circ a_A^{-1}$ and $a_C \circ g \circ a_A^{-1}$. These remain distinct, so it suffices to construct the arrows for the negation normal forms A°, C°. If $i^\circ : \ell \wedge \bot \to A^\circ$ and $r^\circ : C^\circ \to \ell \vee \top$ satisfy the equations, then $i = a_A^{-1} \circ i^\circ : \ell \wedge \bot \to A$ and $r = r^\circ \circ a_C : C \to \ell \vee \top$ satisfy them for the original formulas. We may therefore assume that A, C are in negation normal form.

[3] Lean: `Presentation.occurrence_substitute_expanded`; the composition step is `Relations.Fiber.lift_composition`.

[4] Lean: `Formula.subst_identity`, `Formula.subst_composition`, and `Presentation.substitute_identity`, `Presentation.substitute_composition`, `Presentation.substitute_ext`.

Faithfulness of G gives a pair belonging to exactly one of Gf, Gg. Interchange f, g if necessary, and choose $(x, y) \in Gf \setminus Gg$. The occurrences x and y have the same literal ℓ. Set $D = \ell \wedge \bot$ and $E = \ell \vee \top$. For every formula V, the arrows

$$b_V = \kappa_V \circ \pi_2^{\ell,\bot} : D \to V, \qquad t_V = \iota_2^{\ell,\top} \circ !_V : V \to E$$

have empty occurrence relations.

Construct $i : D \to A$ by induction on A, following the subformula that contains x at each binary connective. At the selected literal, use $\pi_1^{\ell,\bot}$. At a conjunction, pair the recursively constructed arrow with b_V for the unselected factor, placing it in the order specified by the tag. At a disjunction, compose the recursively constructed arrow with the injection of the selected summand. The displayed rules for G show inductively that Gi consists of one pair, whose first component is the sole occurrence of D and whose second component is x.

Construct $r : C \to E$ by induction on C, following the subformula that contains y. At the selected literal, use $\iota_1^{\ell,\top}$. At a conjunction, first project to the selected factor and then apply the recursively constructed arrow. At a disjunction, copair that arrow with t_V on the unselected summand. Thus Gr consists of one pair, whose first component is y and whose second component is the sole occurrence of E.

Relational composition now makes $G(r \circ f \circ i)$ the singleton relation from $\mathrm{Occ}(D)$ to $\mathrm{Occ}(E)$, whereas $G(r \circ g \circ i)$ is empty. Boolean Coherence gives the asserted equations. □

The equations (K_A) pass to retracts. Let θ be a proof congruence on B, and suppose that $i : A \to B$ and $r : B \to A$ satisfy $r \circ i = 1_A$. If (K_B) holds modulo θ, then (K_A) holds modulo θ. Precomposing (K_B) with $i \wedge 1_\bot$ and postcomposing it with $r \vee 1_\top$ gives (K_A) modulo the same proof congruence. Indeed, the left side becomes

$$(r \vee 1_\top) \circ j_B \circ \pi_1^{B,\bot} \circ (i \wedge 1_\bot) = j_A \circ (r \circ i) \circ \pi_1^{A,\bot} = j_A \circ \pi_1^{A,\bot},$$

and zero absorption gives the zero on the right.[5]

Proposition 4.2. *If a substitution congruence θ on B identifies distinct parallel arrows, then (K_A) holds modulo θ for every formula A.*

Proof. Lemma 4.1 and compatibility with composition give (K_A) for a literal ℓ. If $\ell = p$, let σ substitute A for p and fix the other variables. This yields (K_A) for A: under (4), the literal occurrence in each endpoint becomes the block $\mathrm{Occ}(A)$, and the arrow operations are preserved by F_σ; see also Appendix A.3.

If $\ell = \neg p$, substitute $\neg A$ for p, obtaining the equation for $\neg\neg A$. Precompose with $n_A^{-1} \wedge 1_\bot$ and postcompose with $n_A \vee 1_\top$, where $n_A : \neg\neg A \to A$. Put $k_A = j_A \circ \pi_1^{A,\bot} : A \wedge \bot \to A \vee \top$. The product and coproduct equations give the commutative square

$$\begin{array}{ccc} A \wedge \bot & \xrightarrow{k_A} & A \vee \top \\ {\scriptstyle n_A^{-1} \wedge 1_\bot}\downarrow & & \uparrow{\scriptstyle n_A \vee 1_\top} \\ \neg\neg A \wedge \bot & \xrightarrow[k_{\neg\neg A}]{} & \neg\neg A \vee \top. \end{array}$$

Precomposition and postcomposition of the zero arrow give a zero arrow by absorption. The equation for $\neg\neg A$ therefore gives (K_A) for A. □

[5] Lean: `ContextualCollapse.context_retract`.

5 Equality of the product projections

The next lemma identifies two isomorphisms of the types listed in [1, §12.5, p. 294].

Lemma 5.1. *For every formula A, the arrows*

$$q_A = [j_A \circ \pi_1^{A,\bot}, \iota_2^{A,\top}] : (A \wedge \bot) \vee \top \longrightarrow A \vee \top,$$
$$e_A = 1_A \wedge j_A : A \wedge A \longrightarrow A \wedge (A \vee \top)$$

are isomorphisms in B.

Proof. First take $A = p \in P$. By [1, §12.5, p. 294], the source and target of q_p are isomorphic. Each has one occurrence of p, so the relation of any such isomorphism is the unique bijection between these occurrence sets. The arrow q_p has the same relation. Boolean Coherence therefore identifies q_p with that isomorphism.

The same cited list gives an isomorphism $b : p \wedge p \to p \wedge (p \vee \top)$: take $n = m = 1$ in $p^n \wedge (p^m \vee \top) \cong p^{n+m}$ and choose $p^2 = p \wedge p$. The relation Gb is a bijection between two-element occurrence sets. After precomposing b with the product symmetry if necessary, it agrees with the order-preserving bijection Ge_p. Coherence then identifies e_p with an isomorphism. These two identifications also follow from the occurrence-bijection construction in the proof of [3, Lemma 7].

For arbitrary A, substitute A for p and fix the other variables. The functor F_σ sends q_p, e_p to q_A, e_A and preserves their inverse equations. □

Appendix E constructs inverses for these arrows and verifies all four inverse equations.[6]

Lemma 5.2. *Let θ be a proof congruence on* B *such that (K_A) holds modulo θ for every formula A. Then* B/θ *is a preorder.*

Proof. Write $\equiv$ for equality modulo θ. The quotient retains products and coproducts: their pairing, copairing, and defining equations pass to equivalence classes. Fix A, and put $E = A \vee \top$, $S = (A \wedge \bot) \vee \top$, and $s = \iota_2^{A,\top} : \top \to E$. The arrows $0_{A\wedge\bot,E}$ and $s \circ !_{A\wedge\bot}$ have empty occurrence relations and hence are equal in B. Thus (K_A) and the coproduct equations give

$$q_A \equiv [s \circ !_{A\wedge\bot}, s] = s \circ !_S.$$

The inverse equations for q_A hold in the quotient, so

$$1_E = q_A \circ q_A^{-1} \equiv s \circ !_S \circ q_A^{-1} = s \circ !_E.$$

Together with $!_E \circ s = 1_\top$, this makes E terminal in the quotient.

Consequently the projection $r = \pi_1^{A,E} : A \wedge E \to A$ is invertible, with inverse $\langle 1_A, s \circ !_A \rangle$. Lemma 5.1 makes e_A invertible too, and the product equations give $r \circ e_A = \pi_1^{A,A}$. Hence $\pi_1^{A,A}$ is invertible. Its right inverse is the diagonal $\Delta_A = \langle 1_A, 1_A \rangle : A \to A \wedge A$, which must therefore be its inverse. It follows that

$$\pi_2^{A,A} \equiv \pi_2^{A,A} \circ \Delta_A \circ \pi_1^{A,A} = \pi_1^{A,A}.$$

For parallel arrows $f, g : X \to A$, pairing gives $f = \pi_1^{A,A} \circ \langle f, g \rangle \equiv \pi_2^{A,A} \circ \langle f, g \rangle = g$. Thus the quotient is a preorder. □

[6] Lean: `Presentation.ShortProof.qInverseTerm` and `Presentation.ShortProof.eInverseTerm`.

Proof of Theorem 2.1. Suppose that a substitution congruence θ on $\mathsf{B}(P)$ identifies two distinct parallel arrows. Proposition 4.2 gives (K_A) modulo θ for every formula A. Lemma 5.2 then makes $\mathsf{B}(P)/\theta$ a preorder. □

We also obtain maximality in the sense of [1, §9.3, p. 194]. An *interpretation* in a zero-identity Boolean category $\mathcal{C}$ assigns an object of $\mathcal{C}$ to each propositional variable. For an assignment $v : P \to \mathrm{Ob}(\mathcal{C})$, the induced functor $I_v : \mathsf{B}(P) \to \mathcal{C}$ preserves the specified products, coproducts, units, dissociativity arrows, negation maps, η and ϵ.

Corollary 5.3. *Let P be a nonempty set of propositional variables, and let $f, g : A \to C$ be distinct arrows of $\mathsf{B}(P)$. If a zero-identity Boolean category $\mathcal{C}$ satisfies $f = g$ under every interpretation of the propositional variables, then $\mathcal{C}$ is a preorder.*

Proof. Declare two parallel arrows of $\mathsf{B}(P)$ equivalent when all their interpretations in $\mathcal{C}$ agree. Since every I_v preserves the arrow operations, this relation is a proof congruence. For a substitution $\sigma : P \to \mathrm{Form}(P)$, the composite $I_v \circ F_\sigma$ is the interpretation induced by $p \mapsto I_v(\sigma(p))$. The congruence is therefore preserved by substitution. It contains $f = g$, so Theorem 2.1 makes the quotient a preorder. Hence the congruence is total. In particular, $\pi_1^{p,p} = \pi_2^{p,p}$ holds under every interpretation, for any $p \in P$. Interpret p as an arbitrary object X of $\mathcal{C}$. Its two projections $X \wedge X \to X$ are equal. For any arrows $a, b : Y \to X$, composing $\langle a, b\rangle : Y \to X \wedge X$ with the equal projections gives $a = b$. □

For zero-identity distributive lattice categories, [1, §12.5, p. 292] proves maximality without the truth constants and relative maximality when both constants are present. For lattice and dicartesian categories without an assumption of distribution, the maximality and relative-maximality results are proved in [10, §§8–10].

6 A second proof and the Lean formalization

The inverse terms in Appendix E and the coherence argument in Appendix D.5 are taken from the Lean development. Appendix D.2 explains how the declarations and proof steps express the mathematical statements used here. The footnotes identify the source declarations.

In the Lean 4 development [11], formulas and arrow terms are inductive types, with arrow terms indexed by their source and target. The relation `Derives` is generated by the presentation's equations and congruence rules. Its quotient is the category $\mathsf{B}(P)$. Substitution, normalization and the occurrence interpretation are defined on terms and proved to respect `Derives`.

The theorem `relative_maximality` states that a substitution congruence identifying distinct parallel arrows is total. It assumes the arrow operations, a faithful occurrence interpretation, isomorphisms to negation normal forms, preservation of literals along occurrence relations, uniform substitution and double negation, with their equations. Its proof that the quotient is a preorder uses the arrow

$$h_A = [1_A, 1_A] \circ m_{A,A} : A \wedge A \longrightarrow A,$$

where the mix arrow $m_{A,C} : A \wedge C \to A \vee C$ is given, for any formula B, by

$$m_{A,C} = (\pi_1^{A,B} \vee 1_C) \circ d_{A,B,C} \circ (1_A \wedge \iota_2^{B,C}).$$

This is equation (dm) of [1, §11.1, p. 234]. The right-hand side is independent of B. The symmetry of mix gives $h_A \circ \langle \pi_2^{A,A}, \pi_1^{A,A}\rangle = h_A$. Write $\equiv$ for equality modulo a proof congruence containing

(K_A) for every formula. For $B = \bot$, put

$$v_A = d_{A,\bot,A} \circ (1_A \wedge \iota_2^{\bot,A}) : A \wedge A \longrightarrow (A \wedge \bot) \vee A.$$

The coproduct equations and (K_A) give

$$\begin{aligned} j_A \circ h_A &= [j_A \circ \pi_1^{A,\bot}, j_A] \circ v_A \\ &\equiv [0_{A\wedge\bot,A\vee\top}, j_A] \circ v_A = j_A \circ \pi_2^{A,A}. \end{aligned}$$

The last equality holds in B by Boolean Coherence: both occurrence relations send the second copy of $\mathrm{Occ}(A)$ identically to the A-summand of $\mathrm{Occ}(A \vee \top)$ and have no pairs from the first copy. Precomposing with the symmetry of $A \wedge A$ and using the preceding identity for h_A gives $j_A \circ \pi_1^{A,A} \equiv j_A \circ \pi_2^{A,A}$. Precompose this equation with $\langle 1_A, 0_A \rangle$. Since the composite $j_A \circ 0_A$ has empty occurrence relation, coherence gives

$$j_A \equiv \iota_2^{A,\top} \circ !_A.$$

Put $s_A = \iota_2^{A,\top} : \top \to A \vee \top$. The coproduct equations give

$$1_{A\vee\top} = [j_A, s_A] \equiv [s_A \circ !_A, s_A] = s_A \circ !_{A\vee\top}.$$

Since $!_{A\vee\top} \circ s_A = 1_\top$, the object $A \vee \top$ is terminal in the quotient. Composing with the coproduct symmetry gives $\iota_2^{\top,A} \equiv \iota_1^{\top,A} \circ !_A$. For $B = \top$, put $w_A = 1_A \wedge !_A : A \wedge A \to A \wedge \top$. Substituting the preceding equation for $\iota_2^{\top,A}$ in the expression for mix gives

$$\begin{aligned} h_A &\equiv [\pi_1^{A,\top}, 1_A] \circ d_{A,\top,A} \circ (1_A \wedge \iota_1^{\top,A}) \circ w_A \\ &= [\pi_1^{A,\top}, 1_A] \circ \iota_1^{A\wedge\top,A} \circ w_A \\ &= \pi_1^{A,\top} \circ w_A = \pi_1^{A,A}. \end{aligned}$$

The second line uses (1); the third uses the coproduct and product equations. The symmetry of h_A yields $\pi_1^{A,A} \equiv \pi_2^{A,A}$, and pairing equates any two parallel arrows.

For parallel arrows $f, g : A \to C$, define

$$f \cup g = h_C \circ (f \wedge g) \circ \Delta_A : A \longrightarrow C, \qquad \Delta_A = \langle 1_A, 1_A \rangle.$$

This is the union operation of [1, p. 220], written with the product and coproduct operations of B. Its occurrence relation is

$$G(f \cup g) = Gf \cup Gg.$$

Indeed, the diagonal supplies both source copies, $f \wedge g$ acts on the corresponding copies, and h_C joins the two target copies. Faithfulness gives, for parallel $f, g, k : A \to C$,

$$(f \cup g) \cup k = f \cup (g \cup k), \qquad f \cup g = g \cup f, \qquad f \cup f = f.$$

For $a : X \to A$ and $b : C \to Y$, relational composition and faithfulness give

$$\begin{aligned} (f \cup g) \circ a &= (f \circ a) \cup (g \circ a), \\ b \circ (f \cup g) &= (b \circ f) \cup (b \circ g). \end{aligned}$$

On an inhabited hom-set $\mathsf{B}(A, C)$, its zero satisfies $f \cup 0_{A,C} = f$; see also [1, pp. 290–291]. Consequently $\mathsf{B}(A, C)$ is a join semilattice with least element $0_{A,C}$, where

$$f \leq g \iff f \cup g = g \iff Gf \subseteq Gg.$$

Finite unions are independent of the order and repeated summands. On each nonempty hom-set, the empty union is its zero arrow.[7]

Cut elimination and Boolean Coherence are proved for the presentation defining $\mathsf{B}(P)$. Normalization, preservation of literals and uniform substitution on terms and their source and target formulas satisfy the remaining hypotheses of `relative_maximality`. Its application gives `Presentation.maximality`.

The theorem `Presentation.short_maximality` formalizes the proof in Section 5, using the inverse terms for q_A and e_A. The model corollary is proved using interpretations preserving the specified operations and their composition with substitution. The formalization also verifies the occurrence maps and finite-ordinal constructions of Appendix A, the diagram identities of Appendix B, and the cut reductions and rank comparisons discussed in Appendix C.

Five declarations are registered in the Palomar Registry [12]: the two maximality proofs, the congruence dichotomy, nonzero variable identities, and the absence of arrows from $\top$ to $\bot$. Appendix D gives their statements in mathematical notation and relates the arrow terms to arrows of $\mathsf{B}(P)$. It also describes the construction of the coherence proof.

A Substitution in occurrence coordinates

Došen and Petrić's generalized composition of definable connectives [1, §2.5, pp. 45–46] substitutes formula patterns and computes the map labelling the resulting occurrences. We describe this map by the tagged coordinates of [7, §§9.2–9.3]. Each occurrence after substitution is indexed by an occurrence of the outer pattern and an occurrence of the pattern inserted there.

A.1 Occurrence maps and their fibres

For an integer $n \geq 0$, put $\underline{n} = \{1, \ldots, n\}$, with $\underline{0} = \varnothing$. A *formula pattern* on a finite set of placeholders is a formula in those placeholders. Its *occurrence map* records the index of the placeholder at each position, after enumerating the placeholders and reading positions from left to right.

Let $U = \{p_1, \ldots, p_k\}$ consist of distinct placeholders and let $M \in \mathrm{Form}(U)$ have m occurrences. Its occurrence map is

$$o_M : \underline{m} \longrightarrow \underline{k}, \qquad o_M(r) = i \quad \text{when position } r \text{ contains } p_i.$$

For each $i \in \underline{k}$, choose a set $Q_i = \{q_{i,1}, \ldots, q_{i,\ell_i}\}$ of distinct local placeholders and a pattern $N_i \in \mathrm{Form}(Q_i)$ with n_i occurrences. The sets Q_i are pairwise disjoint. Write $o_i : \underline{n_i} \to \underline{\ell_i}$ for the occurrence map of N_i. Empty placeholder sets and empty occurrence sets are allowed; a pattern over an empty placeholder set has no variable occurrences.

Define the sets of occurrence coordinates and placeholder indices by

$$D = \{(r, t) : r \in \underline{m},\ t \in \underline{n_{o_M(r)}}\},$$
$$C = \{(i, s) : i \in \underline{k},\ s \in \underline{\ell_i}\}.$$

Thus (r, t) specifies an occurrence in the copy of $N_{o_M(r)}$ inserted at position r of M. The pair (i, s) specifies a local placeholder of N_i. Define the injections

$$a_r : \underline{n_{o_M(r)}} \longrightarrow D, \quad a_r(t) = (r, t), \qquad b_i : \underline{\ell_i} \longrightarrow C, \quad b_i(s) = (i, s).$$

[7] The Lean laws are `Reduced.union_assoc`, `Reduced.union_comm`, `Reduced.union_idem`, `Reduced.union_pre`, `Reduced.union_post`, and `Reduced.finiteUnion.eq_of_members`. The maps `Reduced.reduceHom` and `Reduced.recoverHom` identify these expressions with arrows of $\mathsf{B}(P)$ and are inverse on each hom-set.

There is a unique function $\gamma : D \to C$ for which, for every $r \in \underline{m}$, the following square commutes:

$$\begin{array}{ccc} \underline{n_{o_M(r)}} & \xrightarrow{a_r} & D \\ {\scriptstyle o_{o_M(r)}}\downarrow & & \downarrow{\scriptstyle \gamma} \\ \underline{\ell_{o_M(r)}} & \xrightarrow[b_{o_M(r)}]{} & C. \end{array} \tag{6}$$

Every element of D is $a_r(t)$ for a unique pair (r,t). Thus these squares determine γ uniquely by

$$\gamma(r,t) = \big(o_M(r), o_{o_M(r)}(t)\big).$$

Define the projections $p_D : D \to \underline{m}$ and $p_C : C \to \underline{k}$ by $p_D(r,t) = r$ and $p_C(i,s) = i$. Then $p_C \circ \gamma = o_M \circ p_D$, so the following square commutes:

$$\begin{array}{ccc} D & \xrightarrow{\gamma} & C \\ {\scriptstyle p_D}\downarrow & & \downarrow{\scriptstyle p_C} \\ \underline{m} & \xrightarrow[o_M]{} & \underline{k}. \end{array} \tag{7}$$

Both composites send (r,t) to $o_M(r)$. If $r \neq r'$ but $o_M(r) = o_M(r') = i$, the copies of $\underline{n_i}$ indexed by r and r' are disjoint subsets of D.

To identify this construction with pattern substitution, replace each $q_{i,s}$ in N_i by (i,s) and denote the resulting pattern over C by $\widetilde{N}_i$. Let $\Phi_0 : U \to \mathrm{Form}(C)$ be given by $\Phi_0(p_i) = \widetilde{N}_i$. Every occurrence of $M[\Phi_0]$ has unique coordinates $(r,t) \in D$. This gives a bijection $\chi : \mathrm{Occ}(M[\Phi_0]) \to D$, and its relation to the label map is the commutative square

$$\begin{array}{ccc} \mathrm{Occ}(M[\Phi_0]) & \xrightarrow[\cong]{\chi} & D \\ {\scriptstyle \lambda_{M[\Phi_0]}}\downarrow & & \downarrow{\scriptstyle \gamma} \\ C & \xrightarrow[1_C]{} & C. \end{array} \tag{8}$$

The occurrence with coordinates (r,t) contains the placeholder $\big(o_M(r), o_{o_M(r)}(t)\big)$.

A.2 The finite-ordinal presentation

The finite ordinals used in [1] number the elements of each block consecutively. Put

$$s_0 = t_0 = 0, \qquad s_r = \sum_{q=1}^{r} n_{o_M(q)} \quad (1 \le r \le m), \qquad t_i = \sum_{j=1}^{i} \ell_j \quad (1 \le i \le k).$$

Numbering each disjoint union consecutively gives bijections

$$\theta_D : D \longrightarrow \underline{s_m}, \quad \theta_D(r,t) = s_{r-1} + t,$$
$$\theta_C : C \longrightarrow \underline{t_k}, \quad \theta_C(i,s) = t_{i-1} + s.$$

Define $\delta : \underline{s_m} \to \underline{t_k}$ as the unique function making the following square commute:

$$\begin{array}{ccc} D & \xrightarrow{\gamma} & C \\ {\scriptstyle \cong}\downarrow{\scriptstyle \theta_D} & & {\scriptstyle \cong}\downarrow{\scriptstyle \theta_C} \\ \underline{s_m} & \xrightarrow[\delta]{} & \underline{t_k}. \end{array} \tag{9}$$

Equivalently, $\delta = \theta_C \circ \gamma \circ \theta_D^{-1}$. For each $(r,t) \in D$, this gives

$$\delta(s_{r-1} + t) = t_{o_M(r)-1} + o_{o_M(r)}(t) \qquad ((r,t) \in D). \tag{10}$$

Every source position belongs to exactly one nonempty block, so this formula determines δ even when some blocks are empty.

In Došen and Petrić's notation, μ and ν_i correspond to o_M and o_i, and $\mu(\nu_1, \ldots, \nu_k)$ corresponds to δ. For $x \in \underline{s_m}$, let r be the unique index with $s_{r-1} < x \leq s_r$. Substituting $t = x - s_{r-1}$ in (10) gives their formula [1, §2.5, p. 46]. The square therefore describes the function on finite ordinals in their generalized composition. The *skeleton* of a pattern is obtained by replacing every placeholder by one fixed symbol and leaving the constants and connectives unchanged. A skeleton and an occurrence map recover the pattern by assigning the indicated placeholder to each position. The generalized composite is therefore determined by the skeleton of $M[\Phi_0]$ and the map δ.

For example, take

$$M(p,q) = q \wedge (p \vee q), \qquad N_1(a,b) = a \wedge b, \qquad N_2(c,d) = d \wedge (c \vee d).$$

The occurrence maps, listed by successive values, are $o_M = (2,1,2)$, $o_1 = (1,2)$, and $o_2 = (2,1,2)$. The source blocks have sizes $3, 2, 3$, while the target blocks list (a,b) and (c,d). Thus

source position	1	2	3	4	5	6	7	8
placeholder	d	c	d	a	b	d	c	d
δ	4	3	4	1	2	4	3	4

For instance, source position 2 has coordinates $(1,2)$. The function γ gives $\gamma(1,2) = (2,1)$, and $\theta_C(2,1) = 3$. Thus $\delta(2) = 3$.

A.3 Shared variables and the maximality argument

The local placeholders may be assigned names in a common variable set Q. Let $\rho : C \to Q$ be that assignment; different elements of C may have the same image. Replace each placeholder $c \in C$ by the variable $\rho(c)$, obtaining the renaming map $(-)[\rho] : \mathrm{Form}(C) \to \mathrm{Form}(Q)$. Define $\sigma : U \to \mathrm{Form}(Q)$ by $\sigma(p_i) = \widetilde{N}_i[\rho]$. Then the following square commutes:

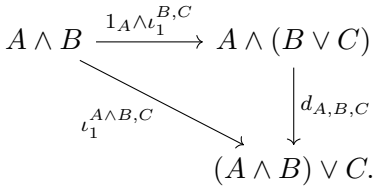
$$\begin{array}{ccc} \mathrm{Form}(U) & \xrightarrow{(-)[\Phi_0]} & \mathrm{Form}(C) \\ {\scriptstyle (-)[\sigma]}\downarrow & & \downarrow{\scriptstyle (-)[\rho]} \\ \mathrm{Form}(Q) & \xrightarrow[1]{} & \mathrm{Form}(Q). \end{array} \tag{11}$$

At p_i, both routes give $\sigma(p_i)$. The constant and connective clauses agree, so induction on the formula proves commutativity. Renaming changes the labels on the occurrences. Thus the same variable of Q may occur in several inserted patterns while the pairs (r,t) distinguish its occurrences.

If the literal isolated in Proposition 4.2 is p, equation (K_A) has source $p \wedge \bot$ and target $p \vee \top$. Each contains one occurrence of p. Substituting A for p therefore gives one block $\mathrm{Occ}(A)$ at each endpoint, and the substituted arrows have type $A \wedge \bot \to A \vee \top$. If the isolated literal is $\neg p$, the substitution $p \mapsto \neg A$ instead gives $\neg\neg A \wedge \bot \to \neg\neg A \vee \top$; the double-negation square in Proposition 4.2 then gives the equation for A.

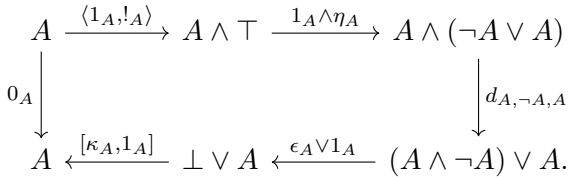
B Diagrams for three defining equations

The first-injection equation. For formulas A, B, C, equation (1) is the commutative triangle



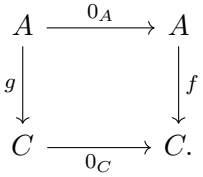
Both paths have source $A \wedge B$ and target $(A \wedge B) \vee C$. The upper arrow is induced by $1_A : A \to A$ and $\iota_1^{B,C} : B \to B \vee C$.

The zero identity. For a formula A, equation (2) is expressed by the commutative diagram



The path across the top, down the right side, and back along the bottom is the five-arrow composite in (2). Its first arrow pairs $1_A : A \to A$ with $!_A : A \to \top$; its last arrow copairs $\kappa_A : \bot \to A$ with $1_A : A \to A$. The arrows between them use $\eta_A : \top \to \neg A \vee A$ and $\epsilon_A : A \wedge \neg A \to \bot$, together with the dissociativity arrow.

The zero law. For any parallel arrows $f, g : A \to C$, equation (3) is the commutative square



The upper-right path is $f \circ 0_A : A \to C$, and the lower-left path is $0_C \circ g : A \to C$. The arrows f and g need not be equal.

C Corrections to the cut-elimination argument used in Boolean Coherence

Došen and Petrić's proof of Boolean Coherence [1, §14.2] uses the cut-elimination theorem for distributive lattice categories proved in [1, §11.2]. Our maximality proof uses Boolean Coherence as Theorem 3.1. We correct two typographical errors in the rank comparisons of that cut-elimination argument. The displayed cut reductions are unchanged.

Write L, R for the left and right ranks of a cut and $r = L + R$ for its total rank. Each one-sided rank counts the nodes of the corresponding cut-formula cluster, so $L, R \geq 1$. The induction orders (d, r) lexicographically, where d is the degree of the cut formula [1, pp. 249,253]. In the following cases, the cut formula is unchanged.

Page 260, case $(w\,\gamma\,3)$**.** The inequality $r < r'$ following the displayed reduction, where r' is the rank of a recursive cut, should read $r' < r$. The two recursive cuts retain the left premise h. Their right cut-formula clusters lie properly inside the original right cluster. If their ranks are R_f, R_g, then $R_f < R$ and $R_g < R$. Thus both recursive cuts have smaller total rank: $L + R_f < r$ and $L + R_g < r$.

Page 262, case $(w \vee 1)$**.** The phrase "whose rank is 1" should read "whose right rank is 1", as in the analogous case on p. 261. After eliminating the first recursive cut, the selected disjunction in the new right premise is introduced by $\vee L$, so its cluster has one node. The left premise is still h. The new cut therefore has total rank $L + 1$, not 1. The original right cluster passes through the contraction and has $R > 1$. Hence $2 \leq L + 1 < L + R = r$, which justifies the second application of the induction hypothesis.

D The registered Lean theorems

The five declarations in the namespace `DosenPetric.Palomar` are stated for arrow terms. We give their statements and relate them to arrows of $\mathsf{B}(P)$. The types are specified in `Challenge.lean`; `Solution.lean` imports their proofs from `DosenPetric.PalomarMaximality`.

D.1 Arrow terms and equations

Let P be a set of propositional variables, and let $\mathcal{T}_P(A, B)$ be the set of arrow terms with source $A \in \mathrm{Form}(P)$ and target $B \in \mathrm{Form}(P)$. Write $\mathcal{E}_{A,B}$ for the relation denoted by `Derives`. It is generated by the defining equations of the presentation, equivalence, composition, conjunction and disjunction. Thus

$$\mathsf{B}(P)(A, B) = \mathcal{T}_P(A, B)/\mathcal{E}_{A,B}.$$

Write $[f]$ for the class of $f \in \mathcal{T}_P(A, B)$. Then $[f] = [g]$ if and only if $f\,\mathcal{E}_{A,B}\,g$. The terms $\mathrm{id}_A, z_A \in \mathcal{T}_P(A, A)$ represent 1_A and the primitive zero endomorphism, respectively; $[z_A]$ is the arrow 0_A of (2). Composition of terms is written $g \circ f$, with f applied first; in Lean this term is `Term.comp f g`.

For a set Q and $\sigma : P \to \mathrm{Form}(Q)$, substitution gives

$$\mathcal{T}_P(A, B) \longrightarrow \mathcal{T}_Q(A[\sigma], B[\sigma]), \qquad f \longmapsto f[\sigma].$$

Induction on an equational derivation shows that $f\,\mathcal{E}_{A,B}\,g$ implies $f[\sigma]\,\mathcal{E}_{A[\sigma],B[\sigma]}\,g[\sigma]$, where the latter relation is formed over Q. Consequently the following square commutes:

$$\begin{array}{ccc} \mathcal{T}_P(A,B) & \xrightarrow{\;f\mapsto[f]\;} & \mathsf{B}(P)(A,B) \\ {\scriptstyle f\mapsto f[\sigma]}\downarrow & & \downarrow{\scriptstyle [f]\mapsto F_\sigma([f])} \\ \mathcal{T}_Q(A[\sigma],B[\sigma]) & \xrightarrow[u\mapsto[u]]{} & \mathsf{B}(Q)(A[\sigma],B[\sigma]). \end{array}$$

A `PrimitiveCongruence P` is a family of equivalence relations $R_{A,B}$ on $\mathcal{T}_P(A,B)$ that contains every defining equation and is preserved by composition, conjunction and disjunction. Explicitly, composition requires

$$f\,R_{A,B}\,f',\quad g\,R_{B,C}\,g' \quad\Longrightarrow\quad g\circ f\,R_{A,C}\,g'\circ f',$$

and, for $\star\in\{\wedge,\vee\}$, the binary operations require

$$f\,R_{A,B}\,f',\quad g\,R_{C,D}\,g' \quad\Longrightarrow\quad f\star g\,R_{A\star C,B\star D}\,f'\star g'.$$

The property `R.Closed` is preservation under every $\sigma : P\to \mathrm{Form}(P)$:

$$f\,R_{A,B}\,g \quad\Longrightarrow\quad f[\sigma]\,R_{A[\sigma],B[\sigma]}\,g[\sigma].$$

The property `R.Total` means

$$\text{for every } A,B \text{ and every } f,g\in\mathcal{T}_P(A,B),\qquad f\,R_{A,B}\,g.$$

These quantifiers do not assert that $\mathcal{T}_P(A,B)$ is nonempty.

D.2 Reading a Lean declaration

The Lean identifiers below refer to the registered source archive [12]. Lean represents a mathematical set by a type and a proposition by an expression of type `Prop`. A proof of a proposition is a term having that proposition as its type. Thus `f : Term P A B` declares an arrow term with source A and target B, whereas `h : Derives f g` declares a proof that the two terms are related by the presentation's equations. The notation `Hom A B` denotes the quotient arrows, rather than the arrow terms.[8]

For example, the statement of `Palomar.maximality`, writing `Not` for the notation $\neg$, is

```
theorem maximality [Nonempty P]
    (R : PrimitiveCongruence P) (closed : R.Closed)
    {A B : Formula P} (f g : Term P A B)
    (hne : Not (Derives f g)) (hfg : R.Rel f g) :
    R.Total
```

Here P is a parameter declared in the surrounding scope as `P : Type u`. This represents the variable set at universe level `u`; the theorem is uniform in that level. The bracketed `[Nonempty P]` is a hypothesis that P is nonempty. The braces in `{A B : Formula P}` mark universally quantified

[8] For the general syntax, see *Theorem Proving in Lean 4*, Chapters 2–3 and 5, https://lean-lang.org/theorem_proving_in_lean4/.

arguments that Lean infers from the types of `f` and `g`. The names `closed`, `hne` and `hfg` name proofs of the three displayed hypotheses; the expression after the last colon is the conclusion.

Dot notation has two uses here. In `Palomar.PrimitiveCongruence`, it gives the namespace of a definition. In `R.Rel`, it selects the relation belonging to R. The expression `R.Closed` applies the defined property `PrimitiveCongruence.Closed` to R. The assertion `R.Total` expands to

```
forall {C D} (k l : Term P C D), R.Rel k l
```

which is the universal statement in the preceding subsection. Consequently the theorem takes the given relation and hypotheses to a proof that every pair of parallel terms is related.

In a proof introduced by `by`, the commands construct this proof term. The command `intro` introduces a universally quantified variable or a hypothesis; `obtain` extracts witnesses and their properties from an existence statement; `rw` replaces expressions using equations; and `exact` supplies a proof of the statement remaining to be proved. Lean's kernel checks that the resulting term has the theorem's displayed type. In particular, `Term.comp f g` applies `f` first, so it denotes $[g] \circ [f]$ on quotient arrows.

The proof of `Palomar.short_maximality` can be read as the following sequence of mathematical implications.

Lean declaration	*Mathematical step*
`Palomar.PrimitiveCongruence.raw_distinct_iff`	$\neg\mathcal{E}(f,g)$ means $[f] \neq [g]$.
`Isolation.normalized_literal_context`	Lemma 4.1 and compatibility with composition give a literal ℓ and (K_ℓ) modulo the induced congruence.
`ContextualCollapse.all_contexts`	Proposition 4.2 gives (K_A) for every formula A.
`Presentation.ShortProof.total`	Lemma 5.2 identifies arbitrary parallel arrows by the q_A, e_A argument.

In this proof `R.fragment` is the induced relation on quotient arrows, with its composition, pairing and copairing laws. The argument `interpretation P` supplies the occurrence interpretation and its proved faithfulness; `normalization P` supplies the normal-form isomorphisms. The arguments `substitution P` and `doubleNegation P` supply the arrow substitution and double-negation maps used in Proposition 4.2. These parameters supply the structures and equations used in the four steps above.

D.3 The five declarations

`maximality` and `short_maximality`. Each declaration has the following statement. Suppose that P is nonempty, R is as above and `R.Closed` holds. If, for some A, B and $f, g \in \mathcal{T}_P(A,B)$,

$$f \, R_{A,B} \, g \qquad \text{and} \qquad \neg(f \, \mathcal{E}_{A,B} \, g),$$

then `R.Total` holds.

The declaration `maximality` uses the argument with the symmetric arrow h_A in Section 6. The declaration `short_maximality` uses the isomorphisms q_A, e_A in Section 5. Both use the occurrence selection of Lemma 4.1 and substitution as in Proposition 4.2. The relation used in these arguments need only be preserved by composition, pairing and copairing, together with uniform substitution. Pairing and copairing preserve the relation induced by R because they are defined using the diagonal, codiagonal and the two binary arrow operations.

`congruence_dichotomy`. For nonempty P and a family R satisfying `R.Closed`, either

$$\text{for every } A, B \text{ and } f, g \in \mathcal{T}_P(A,B), \qquad f\, R_{A,B}\, g \iff f\, \mathcal{E}_{A,B}\, g,$$

or `R.Total` holds. This is the equality-or-totality formulation of Theorem 2.1. Indeed, induction on `Derives` gives $\mathcal{E}_{A,B} \subseteq R_{A,B}$. If the reverse inclusion fails for some pair, that pair satisfies the hypotheses of `maximality`.

`identity_ne_zero`. For every $p \in P$,

$$\neg(\mathrm{id}_p\ \mathcal{E}_{p,p}\ z_p).$$

Equivalently, $1_p \neq 0_p$ in $\mathsf{B}(P)$. The occurrence relation of 1_p is the diagonal on the singleton $\mathrm{Occ}(p)$, whereas the occurrence relation of 0_p is empty. This is the distinction used in Section 3. The declaration takes p as an argument and makes no separate assumption that P is nonempty.

`no_top_to_bot`. For every set P, including the empty set,

$$\mathcal{T}_P(\top, \bot) = \varnothing.$$

Since every arrow is represented by a term, this is equivalent to $\mathsf{B}(P)(\top, \bot) = \varnothing$, as stated in Section 6. Let $\tau : \mathrm{Form}(P) \to \{0,1\}$ be evaluation by the Boolean operations with $\tau(p) = 1$ for every variable p. Induction on $f \in \mathcal{T}_P(A,B)$ gives $\tau(A) \leq \tau(B)$ in the order $0 < 1$. A term of type $\top \to \bot$ would give $1 \leq 0$, which is impossible.

D.4 Congruences on terms and on arrows

For a family R satisfying the conditions above, $\mathcal{E}_{A,B} \subseteq R_{A,B}$ implies that

$$[f]\, \overline{R}_{A,B}\, [g] \quad \iff \quad f\, R_{A,B}\, g$$

defines an equivalence relation on $\mathsf{B}(P)(A,B)$, independent of representatives. For example, if $f\, \mathcal{E}_{A,B}\, f'$ and $g\, \mathcal{E}_{A,B}\, g'$, then equivalence and $\mathcal{E}_{A,B} \subseteq R_{A,B}$ give $f\, R_{A,B}\, g$ if and only if $f'\, R_{A,B}\, g'$. The relations $\overline{R}_{A,B}$ are preserved by composition, pairing and copairing. If `R.Closed` holds, the substitution square with $Q = P$ shows that they are preserved by F_σ for every $\sigma : P \to \mathrm{Form}(P)$.

The relation $\overline{R}$ is equality on arrows if and only if $R = \mathcal{E}$, and totality of $\overline{R}$ is equivalent to `R.Total`. For nonempty P, the dichotomy therefore makes $\overline{R}$ either equality or the total relation. Both are preserved by every arrow operation, including negation. Thus $\overline{R}$ is a substitution congruence in the sense of Section 2.

Conversely, let P be nonempty and let θ be a substitution congruence on $\mathsf{B}(P)$ in the sense of Section 2. Define

$$f\, R^{\theta}_{A,B}\, g \quad \iff \quad [f]\, \theta_{A,B}\, [g].$$

The family R^θ satisfies the defining conditions of `PrimitiveCongruence P` and `Closed`. Distinct θ-equivalent arrows have representatives that are R^θ-related but not $\mathcal{E}$-related. Either maximality declaration then makes R^θ total, and hence makes θ total. The quotient $\mathsf{B}(P)/\theta$ is therefore a preorder, as stated in Theorem 2.1. The correspondence is proved in `DosenPetric.PalomarCorrespondence`.

The theorem `Presentation.coherence` proves Theorem 3.1, and `PresentationSemantics.model_corollary` proves Corollary 5.3. Both names have prefix `DosenPetric.`

In the namespace `DosenPetric.ContextualCollapse`, `second_block` proves $j_A \circ h_A \equiv j_A \circ \pi_2^{A,A}$, and `fold_top` proves $h_A \equiv \pi_1^{A,A}$, as calculated in Section 6. The intermediate equality making $A \vee \top$ terminal is `terminal_retraction`.

D.5 The proof of coherence

Let $\mathcal{R}(P)$ denote the presentation on negation normal forms described in Section 3. We use the same notation for its products, coproducts, units, associativity and symmetry maps, mix and dissociativity. Write G also for its occurrence interpretation; inclusion into $\mathsf{B}(P)$ preserves each tagged relation. A term of $\mathcal{R}(P)$ is *positive* if it contains no η_p or ϵ_p constructor. An arrow is positive when it has such a representative.

The union equations used in the argument below have derivations from the defining equations of $\mathcal{R}(P)$. Naturality of the diagonal, mix and codiagonal gives compatibility with composition; symmetry and associativity give commutativity and associativity. The equation

$$[1_A, 1_A] \circ m_{A,A} \circ \Delta_A = 1_A$$

gives idempotence, and the zero law and the product, coproduct and dissociativity equations give $f \cup 0 = f$. These derivations do not use faithfulness.[9]

The two canonical comparison arrows

$$c^{\wedge}_{X,U,V} = [1_X \wedge \iota_1^{U,V}, 1_X \wedge \iota_2^{U,V}] : (X \wedge U) \vee (X \wedge V) \longrightarrow X \wedge (U \vee V),$$
$$c^{\vee}_{U,V,X} = \langle \pi_1^{U,V} \vee 1_X, \pi_2^{U,V} \vee 1_X \rangle : (U \wedge V) \vee X \longrightarrow (U \vee X) \wedge (V \vee X)$$

have, respectively, a section $s_{X,U,V}$ and a retraction $t_{U,V,X}$:

$$c^{\wedge}_{X,U,V} \circ s_{X,U,V} = 1_{X \wedge (U \vee V)}, \qquad t_{U,V,X} \circ c^{\vee}_{U,V,X} = 1_{(U \wedge V) \vee X}.$$

Appendix E defines these maps by composites, products and coproducts of identities, associativity, symmetry and dissociativity maps, diagonals and codiagonals. Their two equations follow from the product, coproduct, dissociativity and mix equations.[10]

Iterating the first split comparison gives, for every formula A of $\mathcal{R}(P)$, a disjunctive normal-form expansion D_A and positive arrows

$$i_A : A \longrightarrow D_A, \qquad r_A : D_A \longrightarrow A, \qquad r_A \circ i_A = 1_A.$$

Here D_A is a coproduct of conjunctions of literals and constants. The second split comparison gives a conjunctive expansion C_A, a product of disjunctions of literals and constants, with positive arrows $A \to C_A \to A$ whose composite is 1_A. Induction on A constructs these expansions: conjunctions of disjunctive expansions use s, disjunctions of conjunctive expansions use $c^{\vee}$, and the other connective combines expansions by its bifunctor. Only the displayed retraction equations are needed.[11]

Consider a positive arrow from a conjunction U of literals and constants to a disjunction V of literals and constants. At occurrences $x \in \mathrm{Occ}(U)$ and $y \in \mathrm{Occ}(V)$ of the same literal ℓ, the successive product projections and coproduct injections give

$$p_x : U \longrightarrow \ell, \qquad j_y : \ell \longrightarrow V.$$

The relation of $j_y \circ p_x$ is the singleton $\{(x, y)\}$. Gentzenization, cut elimination and atomization express the given arrow as a finite union of these paths; an empty union uses the zero in its inhabited hom-set. The relation of that union records exactly its paths. Idempotence and commutativity

[9] Lean: `Reduced.union_pre`, `Reduced.union_post`, `Reduced.union_comm`, `Reduced.union_assoc`, `Reduced.union_idem`, and `Reduced.union_id_zero`.

[10] Lean: `Reduced.Gentzen.DistributiveEpi.section_comparison` and `Reduced.Gentzen.DistributiveMono.comparison_retraction`.

[11] Lean: `Reduced.Gentzen.dnf_expansion` and `Reduced.Gentzen.cnf_expansion`.

make repetitions and their order irrelevant, so equal occurrence relations give equal positive arrows at these endpoints.[12] The coproduct and product equations extend this assertion to arrows from a disjunctive expansion to a conjunctive expansion.

For positive $f, g : A \to B$, choose a disjunctive expansion of A and a conjunctive expansion of B. Write their maps as $i_A : A \to D_A$, $r_A : D_A \to A$ and $i_B : B \to C_B$, $r_B : C_B \to B$, with $r_A \circ i_A = 1_A$ and $r_B \circ i_B = 1_B$. If $Gf = Gg$, the arrows $i_B \circ f \circ r_A, i_B \circ g \circ r_A : D_A \to C_B$ have equal relations and hence are equal. Precomposing with i_A and postcomposing with r_B gives $f = g$. This proves positive coherence at arbitrary endpoints.[13]

For arbitrary terms $f, g : A \to B$ of $\mathcal{R}(P)$, atomic factorization constructs formulas A', B', arrows $u : A \to A'$, $v : B' \to B$ and positive terms $f_0, g_0 : A' \to B'$ such that, on quotient arrows,

$$[f] = v \circ [f_0] \circ u, \qquad [g] = v \circ [g_0] \circ u.$$

It also gives injections $a : \mathrm{Occ}(A) \to \mathrm{Occ}(A')$ and $b : \mathrm{Occ}(B) \to \mathrm{Occ}(B')$ with

$$G[f_0] = (a \times b)(G[f]), \qquad G[g_0] = (a \times b)(G[g]).$$

The right sides are direct images of relations. Thus equality of $G[f]$ and $G[g]$ gives equality of the middle relations. Positive coherence equates $[f_0]$ and $[g_0]$, and the factorizations give $[f] = [g]$.[14] The normalization and recovery maps of Section 3 then transfer faithfulness from $\mathcal{R}(P)$ to $\mathsf{B}(P)$.[15]

E Inverses of q_A and e_A

We construct inverses for the arrows in Lemma 5.1. Throughout this appendix,

$$\alpha^{\star}_{X,Y,Z} : (X \star Y) \star Z \longrightarrow X \star (Y \star Z), \qquad \beta^{\star}_{X,Y} : X \star Y \longrightarrow Y \star X \quad (\star \in \{\wedge, \vee\})$$

are associativity and symmetry.

E.1 The inverse of q_A

Put $S_A = (A \wedge \bot) \vee \top$. Define $u_A : A \to S_A$ by the path

$$A \xrightarrow{\langle 1_A, !_A \rangle} A \wedge \top \xrightarrow{1_A \wedge \iota_2^{\bot,\top}} A \wedge (\bot \vee \top) \xrightarrow{d_{A,\bot,\top}} S_A,$$

and put

$$\bar{q}_A = [u_A, \iota_2^{A \wedge \bot, \top}] : A \vee \top \longrightarrow S_A.$$

The occurrence relation of $\bar{q}_A$ is the graph sending $(1, x) \in \mathrm{Occ}(A \vee \top)$ to $(1, (1, x)) \in \mathrm{Occ}(S_A)$; Gq_A is the inverse graph. Boolean Coherence therefore gives

$$\bar{q}_A \circ q_A = 1_{S_A}, \qquad q_A \circ \bar{q}_A = 1_{A \vee \top}.$$

Thus $\bar{q}_A = q_A^{-1}$.[16]

[12] Lean: `Reduced.Gentzen.G.mix_normal_form`, `Reduced.Gentzen.HasPureNormalForm.faithful`, and `Reduced.Gentzen.pure_positive_coherence`.

[13] Lean: `Reduced.Gentzen.represented_coherence` and `Reduced.Gentzen.positive_coherence`.

[14] Lean: `Reduced.commonAtomicFactor` and `Reduced.faithful_of_positive_coherence`.

[15] Lean: `Reduced.boolean_coherence` and `Presentation.coherence`.

[16] Lean: `Presentation.ShortProof.qInverseTerm`, `Presentation.ShortProof.q_qInverse`, and `Presentation.ShortProof.qInverse_q`.

E.2 The inverse of e_A

Define $c_{X,Y;U,V} : (X \wedge Y) \wedge (U \vee V) \to (X \wedge U) \vee (Y \wedge V)$ by the path

$$\begin{array}{c}(X \wedge Y) \wedge (U \vee V)\\ \downarrow{\scriptstyle 1_{X\wedge Y} \wedge \beta^{\vee}_{U,V}}\\ (X \wedge Y) \wedge (V \vee U)\\ \downarrow{\scriptstyle \alpha^{\wedge}_{X,Y,V\vee U}}\\ X \wedge (Y \wedge (V \vee U))\\ \downarrow{\scriptstyle 1_X \wedge d_{Y,V,U}}\\ X \wedge ((Y \wedge V) \vee U)\\ \downarrow{\scriptstyle 1_X \wedge \beta^{\vee}_{Y\wedge V,U}}\\ X \wedge (U \vee (Y \wedge V))\\ \downarrow{\scriptstyle d_{X,U,Y\wedge V}}\\ (X \wedge U) \vee (Y \wedge V).\end{array}$$

Set

$$s_{X,U,V} = c_{X,X;U,V} \circ (\Delta_X \wedge 1_{U\vee V}) : X \wedge (U \vee V) \longrightarrow (X \wedge U) \vee (X \wedge V).$$

With $T_A = A \wedge \top$, define

$$\bar{e}_A = [1_{A\wedge A}, 0_{T_A,A\wedge A}] \circ s_{A,A,\top} : A \wedge (A \vee \top) \longrightarrow A \wedge A.$$

The composite $\Delta_A \circ \pi_1^{A,\top} : T_A \to A \wedge A$ supplies the witness defining this zero. The relation of $s_{A,A,\top}$ copies each occurrence of its first A into both target summands and sends the occurrences of the second A to the second factor of the first summand. Postcomposition with $[1_{A\wedge A}, 0_{T_A,A\wedge A}]$ removes the second summand. Hence $G\bar{e}_A$ sends $(1, x)$ to $(1, x)$ and $(2, (1, x))$ to $(2, x)$. These assignments are inverse to Ge_A. Coherence gives

$$\bar{e}_A \circ e_A = 1_{A\wedge A}, \qquad e_A \circ \bar{e}_A = 1_{A\wedge(A\vee\top)}.$$

Thus $\bar{e}_A = e_A^{-1}$.[17] Both inverse constructions apply to every formula A, including formulas with empty occurrence sets.

E.3 The split comparisons used in coherence

The map $s_{X,U,V}$ is also a section of $c^{\wedge}_{X,U,V}$ in Appendix D.5. To construct the retraction of $c^{\vee}_{U,V,X}$, first put

$$v_{U,X;V} = (\beta^{\wedge}_{V,U} \vee 1_X) \circ d_{V,U,X} \circ \beta^{\wedge}_{U\vee X,V} : (U \vee X) \wedge V \longrightarrow (U \wedge V) \vee X.$$

[17] Lean: `Presentation.ShortProof.sourceConjTerm`, `Presentation.ShortProof.sectionTerm`, `Presentation.ShortProof.eInverseTerm`, `Presentation.ShortProof.e_eInverse`, and `Presentation.ShortProof.eInverse_e`.

For another formula Y, put

$$b_{U,X;V,Y} = \alpha^{\vee}_{U\wedge V,X,Y} \circ (v_{U,X;V} \vee 1_Y) \circ d_{U\vee X,V,Y}$$
$$: (U \vee X) \wedge (V \vee Y) \longrightarrow (U \wedge V) \vee (X \vee Y).$$

Then

$$t_{U,V,X} = (1_{U\wedge V} \vee [1_X, 1_X]) \circ b_{U,X;V,X} : (U \vee X) \wedge (V \vee X) \longrightarrow (U \wedge V) \vee X.$$

Expanding the definitions and using the presentation equations gives

$$\pi_i^{X,U\vee V} \circ c^{\wedge}_{X,U,V} \circ s_{X,U,V} = \pi_i^{X,U\vee V} \quad (i = 1, 2),$$
$$t_{U,V,X} \circ c^{\vee}_{U,V,X} \circ \iota_i^{U\wedge V,X} = \iota_i^{U\wedge V,X} \quad (i = 1, 2).$$

The product and coproduct equations give the two split identities. For the first projection in the first line and the second injection in the second line, the calculation uses $[1_Z, 1_Z] \circ m_{Z,Z} \circ \Delta_Z = 1_Z$. These calculations use only the product, coproduct, associativity, symmetry, dissociativity and mix equations, so the same expressions and identities hold in $\mathcal{R}(P)$.

Use of AI tools

OpenAI's ChatGPT assisted with the development of the proof, preparation of the manuscript, and checks of the arguments and references. OpenAI Codex was used to develop the Lean formalization and to run its verification checks. The author takes responsibility for the mathematical claims and the final text.

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